

Fish Length Estimation with Vision Foundation Models: Can They Beat the AutoFish CNN Baseline?

Abu Bakar · Laksh Jiwani · Shahman Butt

Area Seminar: Deep Learning for Maritime Vision Applications, Visual and Analytic Computing in Ocean Technologies (VACOT), University of Rostock · Supervisor: Bohan Zhuang, M.Sc. · Professor: Stefan Oehmcke

Motivation & Question

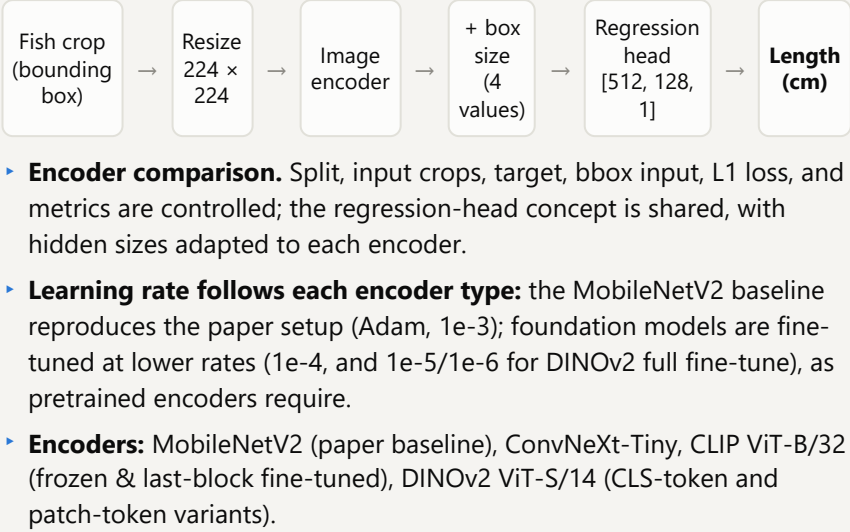
- Fish length is a key measurement for fisheries science and catch monitoring. Manual measuring is slow and invasive.
- The AutoFish benchmark (Bengtson et al., WACVW 2025) provides images, annotations, and a CNN length-regression baseline.
- Vision foundation models promise strong general features, but do they help **precise metric regression**?

Question: Do foundation-model encoders (DINOv2, CLIP) or newer supervised encoders (ConvNeXt) improve fish length estimation over the reproduced AutoFish MobileNetV2 baseline, using identical data, splits, and metrics?

Data & Setup

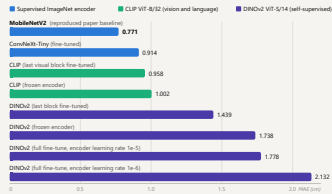
- AutoFish dataset:** 1,500 top-view images, 454 unique fish, 18,157 fish annotations, organized in 25 fish groups.
- Official group-level split:** 15 train / 5 validation / 5 test groups. No fish appears in two splits (one duplicate fish identity was found and removed).
- Two test regimes:** non-occluded (Set1+Set2, fish laid separately) vs. occluded (“All”, fish overlapping).
- Metric:** mean absolute error (MAE) of predicted length in cm, lower is better. RMSE, MAPE, and R^2 are tracked as secondary metrics.

Method / Workflow



Main result: MobileNetV2 is the best single model

Mean absolute error (cm) on the full test set (3,759 fish, occluded + non-occluded). Lower is better.



Takeaway: among single models, every foundation-model variant trails the fine-tuned MobileNetV2 baseline; supervised encoders (blue) top the ranking. Combining models as an ensemble then beats the baseline (see below).

Figure: full-test MAE per model; identical split / input / head / metrics, learning rate set per encoder type (see Method).

Baseline reproduction

Non-occluded test MAE (Set1+Set2), our run vs. the AutoFish paper:

Paper (Bengtson et al.)

0.62 cm

Our reproduction

0.633 cm

A difference of only 0.013 cm, so the baseline and our whole comparison pipeline can be trusted.

Key findings

- Baseline reproduced** within 0.013 cm; best single model is MobileNetV2 (0.771 cm).
- An ensemble beats the baseline** (0.711 cm); no single foundation model does.
- Patch tokens reliably beat CLS** for DINOv2 (3 seeds, controlled).
- Mask segmentation (Mask2Former-style) as a third task; DINOv3 pending weight access.

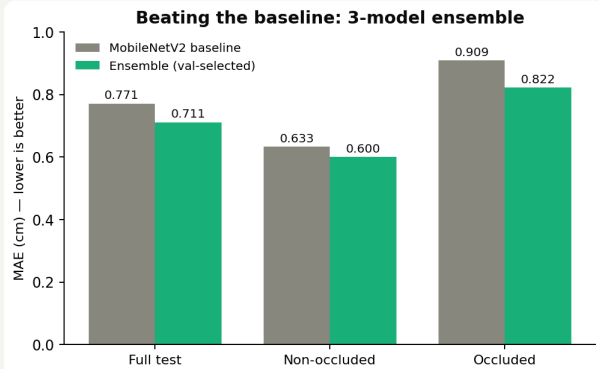
Next steps

- Multi-seed the top length models; try EfficientNet-B0 and patch pooling for CLIP.
- Mask segmentation (Mask2Former-style) as a third task; DINOv3 pending weight access.

Limitations

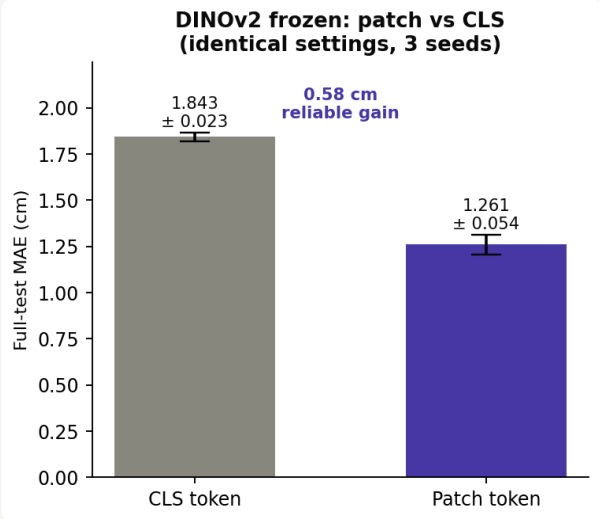
- Single-run for the top length models (multi-seed done for the DINOv2 patch-vs-CLS comparison).
- Foundation models tested at small scale (ViT-S/14, ViT-B/32); larger variants may differ.

We beat the baseline (ensemble)



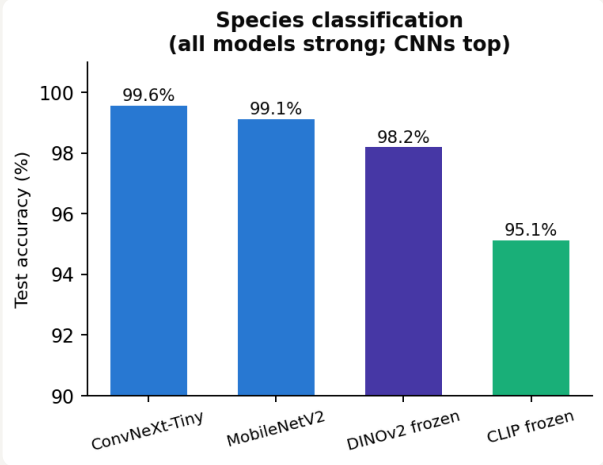
- 0.711 vs 0.771 cm** full-test (~7.8%); wins on both subsets.
- Members chosen on validation, reported once on test (no test-tuning).

Patch tokens beat CLS (DINOv2)



- 1.261 vs 1.843 cm**, 3 seeds, non-overlapping ranges.
- Best DINOv2 setup; last-block fine-tune did not help.

Species classification



- All encoders **95–99.6%**; CNNs top.
- DINOv2 rises from last (length) to near-top (species).